

MAIN-BELT ASTEROID

243 Ida

AVERAGE DISTANCE TO SUN	266 million miles (428 million km)
ORBITAL PERIOD	4.84 years
ROTATION PERIOD	4.63 hours
LENGTH	37 miles (60 km)
DATE OF DISCOVERY	September 29, 1884

Ida was one of 119 asteroids discovered by the Austrian astronomer Johann Palisa, who, together with Max Wolf of Heidelberg, Germany, was a pioneer in the use of photography to produce star maps and hunt for minor planets (another name for asteroids). Ida is a member of the Koronis

family. Asteroidal families were discovered by the Japanese astronomer Hirayama Kiyotsugu in 1918. He found that there were groups of asteroids with very similar orbital parameters. The individual members were strung out on one orbit and formed a stream of minor bodies in the inner solar system (see p.170). Koronis is the most prominent member of Ida's family.

Ida is famous because the Galileo spacecraft imaged it in detail as it flew within 6,800 miles (11,000 km) during August 1993, on its way to Jupiter. Because Ida makes a complete rotation every 4 hours 36 minutes, Galileo was able to image most of the surface during the flyby. Ida was originally thought to be an S-type

DACTYL

At just 1 mile (1.6 km) long, Dactyl is tiny. Its orbit around Ida is nearly circular, with a radius of about 56 miles (90 km) and an orbital period of about 27 hours.

asteroid like Gaspra (see opposite), but observations revealed that its density is too low and it is more likely to be a C-type asteroid. It has about five times more craters per unit area than Gaspra, indicating that its surface is considerably older. The most exciting outcome of the Galileo flyby was the discovery that Ida has its own moon, Dactyl. This binary system is thought to have been formed during the asteroid collision and breakup that created the Koronis family.



IDA AND ITS MOON

Dactyl was the first asteroid satellite to be discovered. Ann Harch, a Galileo mission member, noticed it when examining images that had been stored on the spacecraft when it passed Ida 6 months earlier.